José Fabián Sifuentes Gálvez

Mechatronics Engineer | Fullstack Developer | Embedded Systems & Robotics

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Professional Profile

Mechatronics Engineer (PUCP) and Fullstack Developer with 3+ years of experience spanning the entire technology stack: embedded systems and robotics, PCB electronic design, satellite communications, and end-to-end web application development. Proven ability to integrate hardware and software — from C/C++ firmware and PyQt5 desktop interfaces to React/Node.js web platforms deployed on AWS. Strong systems-level thinking, technical precision, and a track record delivering complex projects in academic, space, and commercial environments.

Experience

Professional Intern — Space Engineering · INRAS – PUCP · pre-professional until 2025 — Jul 2024 – Present

- Designed and developed the graphical interface (PyQt5) for the ground station of the INTISAT satellite mission.
- Developed UHF communication systems and 3D simulations of space tethers.
- Integrated satellite telemetry and real-time data link protocols.

Developer — Embedded Systems & Interfaces · Bear Tech — Jan – Jun 2025

- Built the complete software for an internal pipeline-inspection robot: joystick teleoperation over TCP/IP, simultaneous streaming of 6 cameras (4 in a 360° array), and IMU + odometry navigation with 3D visualization.
- Integrated automatic structural-fault detection with YOLOv8 (7 classes) into the PyQt5 interface.
- Designed multilayer PCB electronics with EasyEDA: schematics, routing, and physical prototype validation.

Fullstack Developer · Freelance / Commercial Project — 2022 – Present

- Developed a project management web application: staff, workflows, and organizational resource allocation.
- Core stack: React (frontend), Node.js + NestJS (backend), Django (APIs), SQL Server, PostgreSQL, MySQL, MariaDB.
- AWS infrastructure: EC2 instances for servers and Lambda functions for serverless processes.
- Professional Git repository management: feature branches, pull requests, and code reviews.

Student Researcher — Bionics · LIBRA – PUCP — Feb – Sep 2024

- Conducted technical analysis and systematic review of upper-limb myoelectric prostheses.
- Participated in the OCTAHAND project for prosthesis recovery, repair, and improvement.

Featured Projects

INTISAT — UHF Communications Payload Lead · Satellite mission, APSCO competition — 2024 – Present

- Ground station interface, telemetry system, and satellite communication systems.

AYNISAT — Peruvian CubeSat (CONIDA) · INRAS — 2024 – Present

- Communications payload with STM32: electronic design in EasyEDA and UHF link testing.

LINKU — Space Stereophotogrammetry · Speaker at UNI Space TechWeek — 2025 – Present

- 3D reconstruction with stereo vision on Raspberry Pi and AI-based neural sensor fusion.

Robotic Teleoperation with Haptic Feedback · personal project — 2025

- Remote robotic arm control with real-time compressed video and bidirectional haptic feedback over Linux.

Technical Skills

Programming: Python, JavaScript (ES6+), TypeScript, C, C++, MATLAB
Web: React, HTML5, CSS3 · Node.js, NestJS, Django, REST APIs
Embedded & Hardware: STM32, Raspberry Pi 5, Arduino · EasyEDA, KiCad (PCB) · I2C, SPI, UART, CAN
Robotics & Simulation: ROS2, Gazebo, RViz, CoppeliaSim · forward/inverse kinematics, robot control
AI & Computer Vision: YOLOv8, neural networks, 3D reconstruction, image processing, sensor fusion
Databases: PostgreSQL, SQL Server, MySQL, MariaDB, Firebase
Cloud & DevOps: AWS (EC2, Lambda), Firebase, Git (branching, PRs, code reviews), Linux/Windows
Mechanical Design: SolidWorks, Fusion 360, Autodesk Inventor, ANSYS

Education, Languages & Certifications

Mechatronics Engineering · Pontificia Universidad Católica del Perú (PUCP) — 2020 – 2025 · Lima, Peru
Languages: Spanish (native) · English — TOEFL (ICPNA)
Certifications: PERUMEC 2025 — robotics competitor · UNI Space TechWeek — speaker (LINKU project)